

Experiment 1

Aim :

The objective of this experiment is to understand and implement basic digital logic circuits using Logisim Evolution. The experiment focuses on designing and simulating fundamental components such as multiplexers, priority encoders, and various combinational logic circuits including AND, OR, NOT, and NOR gates.

Procedure :

In this experiment, multiple digital logic circuits were designed and simulated using Logisim.

- An **8:1 multiplexer** was implemented to select one input from multiple inputs based on select lines.
- A **priority encoder (8:3)** was designed to convert multiple input signals into a binary output representing the highest priority input.
- Basic logic gates such as **NOT, AND, and OR** were implemented to understand individual logical operations.
- A **combinational circuit using NOR gates** was also designed and verified.

All circuits were tested by changing input values and observing the corresponding outputs to ensure correct functionality.

Lab Implementation (Logisim)

The implementation includes:

- 8:1 Multiplexer circuit
- Priority Encoder (8:3)
- NOT gate
- AND gate
- OR gate
- NOR-based combinational circuit

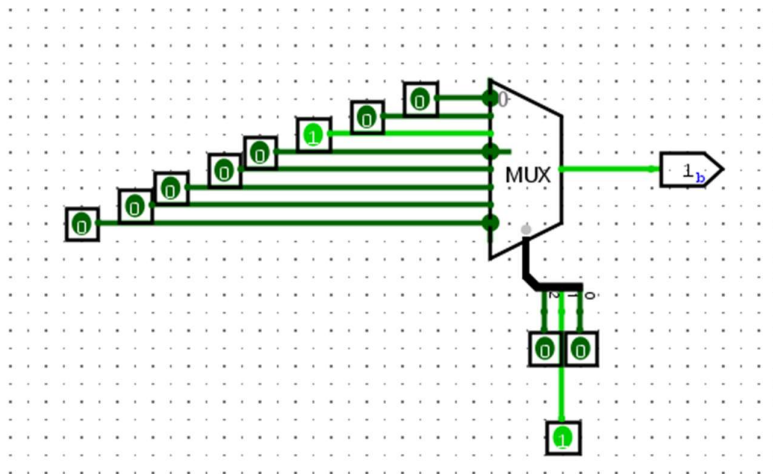
1. 8:1 Multiplexer

An 8-to-1 multiplexer is a combinational circuit that selects one of eight inputs (I0–I7) and forwards it to a single output (Y) based on three selection lines (S2, S1, S0).

Function:

The output depends on the select lines:

- $S_2 S_1 S_0 = 000 \rightarrow Y = I_0$
- $S_2 S_1 S_0 = 001 \rightarrow Y = I_1$
- ...
- $S_2 S_1 S_0 = 111 \rightarrow Y = I_7$



Circuit Diagram of MUX

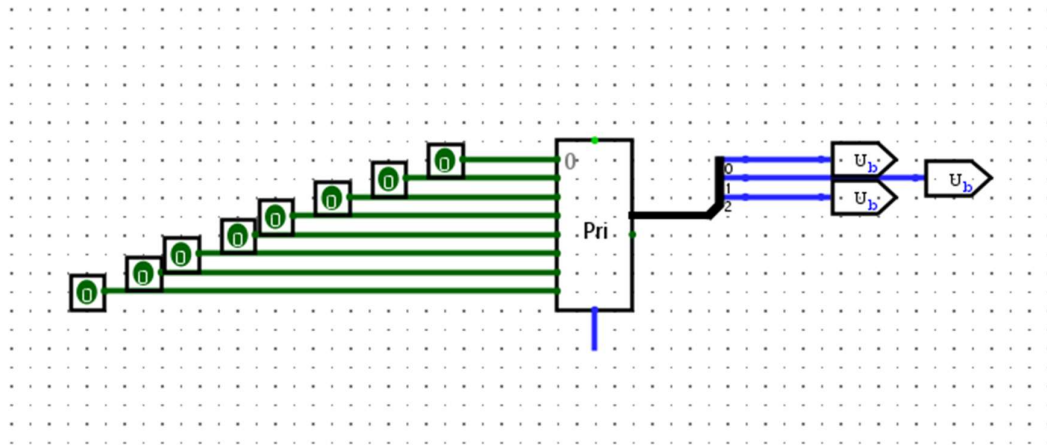
2. Priority Encoder (8:3)

A priority encoder is a combinational circuit that gives output based on the **highest priority active input**.

- It has 8 inputs (D0–D7) and 3 outputs (Q2, Q1, Q0).
- If multiple inputs are HIGH, the encoder selects the highest priority input.

D0	D1	D2	D3	D4	D5	D6	D7	Q1	Q2	Q3
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

Truth table of Encoder(8*3)



Circuit Diagram

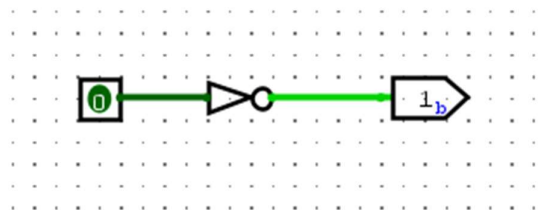
3. NOT Gate

A NOT gate performs logical inversion.

Input Output

0 1

1 0



NOT GATE

4. AND Gate

An AND gate produces output HIGH only when all inputs are HIGH.

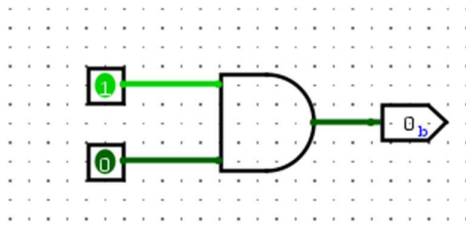
A B Output

0 0 0

0 1 0

1 0 0

1 1 1



AND GATE

5. OR Gate

An OR gate produces output HIGH if at least one input is HIGH.

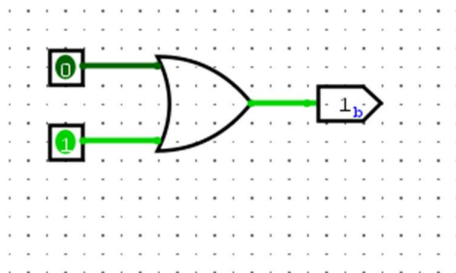
A B Output

0 0 0

0 1 1

1 0 1

1 1 1



OR GATE

6. NOR Gate Circuit

A NOR gate is the combination of OR followed by NOT.

Function:

Output is HIGH only when all inputs are LOW.

A B Output

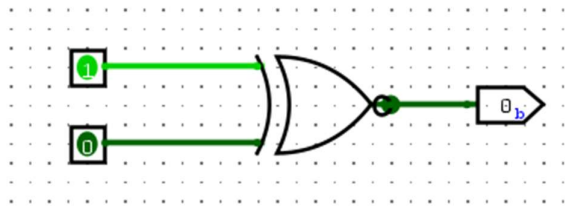
0 0 1

0 1 0

1 0 0

A B Output

1 1 0



NOR GATE

In your circuit (top-right of your image), multiple NOR gates are combined to form a **combinational logic system**, and the final output is verified based on inputs.

Result :

The simulation results showed that:

- The **multiplexer correctly selected inputs** based on select lines.
- The **priority encoder produced correct binary outputs** for the highest priority input.
- The **NOT, AND, and OR gates worked correctly** for all input combinations.
- The **NOR-based circuit produced accurate logical outputs**, confirming proper design.

The experiment was successfully completed, and all circuits were verified using Logisim. This helped in understanding the behavior and implementation of combinational logic circuits.